

Pizza Sales Analysis

Project Overview

The Pizza Sales Analysis project was developed using MySQL and Microsoft Excel to analyze pizza sales performance, customer ordering behavior, and product trends.

The objective of this project is to transform raw transactional data into meaningful business insights through SQL analysis and interactive dashboard visualization.

Business Problem Statement

The business generates large amounts of pizza sales data daily, but without analysis it becomes difficult to understand:

- Customer ordering patterns
- Peak sales periods
- Best-selling products
- Revenue-driving categories
- Product performance trends

The goal of this project is to analyze sales data and create a dashboard that supports data-driven business decision-making.

Project Objectives

- Analyze total revenue and order performance
 - Identify peak ordering days and hours
 - Understand customer preference by pizza category and size
 - Identify best-selling and worst-selling pizzas
 - Build an interactive dashboard for business reporting
 - Generate business recommendations from sales insights
-

Tools & Technologies Used

Tool	Purpose
MySQL	Data analysis and querying
Microsoft Excel	Dashboard creation
SQL	KPI and trend analysis
Pivot Tables & Charts	Data visualization
Slicers	Interactive filtering

Key Business Questions

- What is the total revenue generated from pizza sales?
 - Which pizza categories contribute the highest sales?
 - What are the peak ordering days and hours?
 - Which pizza sizes are most preferred by customers?
 - Which pizzas are best sellers and worst sellers?
 - How can the business improve operational efficiency and sales performance?
-

Database Creation

A dedicated MySQL database named `Pizza_sales_analysis` was created to store and manage pizza sales data.

Table Structure

The `pizza_sales` table was created with the following important fields:

- `pizza_id`
- `order_id`
- `pizza_name`
- `pizza_category`
- `pizza_size`

- quantity
- total_price
- order_date
- order_time

The dataset was imported from a CSV file into MySQL using the Import Wizard.

Project Workflow

Step 1: Data Import

- Imported pizza sales CSV dataset into MySQL
- Verified records and column structure

Step 2: Data Cleaning

- Converted order_date into DATE format
- Validated imported data
- Checked sample records

Step 3: SQL Analysis

Performed KPI and trend analysis using SQL queries:

- Total Revenue
- Average Order Value
- Total Orders
- Total Pizzas Sold
- Daily & Hourly Trends
- Category-wise Analysis
- Best & Worst Sellers

Step 4: Dashboard Creation

Created an interactive Excel dashboard using:

- KPI Cards
- Bar Charts
- Pie Charts

- Line Charts
- Slicers

Step 5: Business Insight Generation

Generated business recommendations based on dashboard analysis.

Dashboard Insights & Business Recommendations

1. Peak Sales Timing

Insight

- Highest order activity occurs on Friday and Saturday
- Peak ordering hours are between 12 PM–1 PM and 5 PM–8 PM

Business Recommendation

Increase staffing, kitchen preparation, and inventory availability during peak hours to improve customer service and reduce delays.

2. Category Performance

Insight

- The Classic category contributes the highest share of total sales and orders.

Business Recommendation

Focus promotional campaigns and combo offers on high-performing categories to maximize revenue generation.

3. Pizza Size Analysis

Insight

- Large pizzas generate the highest sales contribution among all pizza sizes.

Business Recommendation

Introduce family meal deals and upselling strategies to increase average order value.

4. Best-Selling Products

Insight

Top-performing pizzas include:

- Pepperoni Pizza
- Barbecue Chicken Pizza
- California Chicken Pizza

These pizzas consistently attract high customer demand.

Business Recommendation

Feature best-selling pizzas in advertisements, banners, and combo offers to further boost sales.

5. Worst-Selling Products

Insight

Lowest-performing pizzas include:

- Brie Carre Pizza
- Calabrese Pizza
- Mediterranean Pizza

These products contribute the least to total sales.

Business Recommendation

Review pricing strategy, improve menu visibility, or redesign promotional campaigns for underperforming products.

SQL Concepts Used

- CREATE DATABASE
 - CREATE TABLE
 - UPDATE
 - ALTER TABLE
 - STR_TO_DATE()
 - SUM()
 - COUNT()
 - DISTINCT
 - ROUND()
 - GROUP BY
 - ORDER BY
 - WHERE Clause
 - Aggregate Functions
 - Date Functions
-

Conclusion

This project demonstrates how SQL and Excel can be used together for business analysis and dashboard reporting.

The analysis helped identify customer behavior patterns, peak business periods, high-performing products, and revenue-driving categories.

The final dashboard provides a clear and interactive view of business performance, enabling better operational and marketing decisions.

PIZZA



SALE

Total Revenue

\$69,793

Avg Order value

\$37.83

Total Pizzas Sold

4232

Total Orders

1845

Avg Pizzas Per Order

2.29

Peak Days and Hours

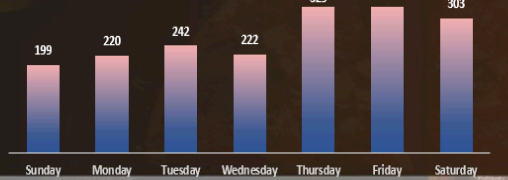
DAYS

Order activity **rises** sharply toward the weekend, especially **Fri-Sat**

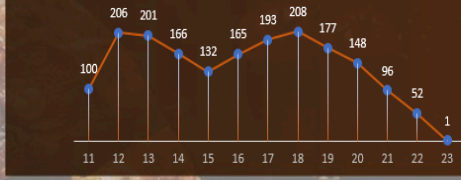
HOURS

Peaks occur around **12-01 PM** and **5-8 PM**

Daily Trend for Total Orders



Hourly Trend for Total Orders



Sales by Category & Size

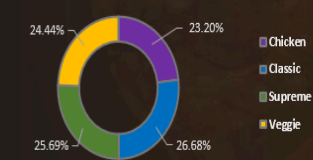
CATEGORY

Classic category **leads** in both sales and total orders

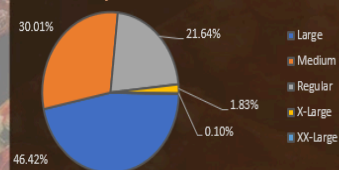
SIZE

Large Size Pizza generate the highest sales & contribution

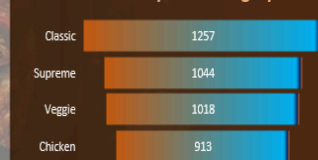
% of Sales by Pizza Category



% of Sales by Pizza Size



Total Pizzas Sold by Pizza Category



Best & Worst Sellers

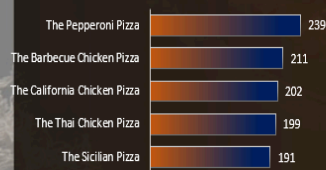
BEST

Classic deluxe & chicken pizzas are the **sellers** and revenue generators

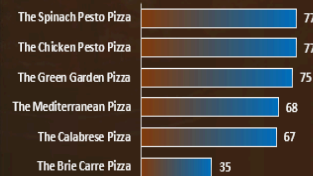
WORST

The Brie Carre Pizza is at **bottom** in both orders and revenue

Top 5 Best Sellers by Total Pizzas Sold



Bottom 5 Worst Sellers by Total Pizzas Sold



order_date

Jan 2015

MONTHS

2015

JAN

FEB

MAR

APR

MAY

JUN

JUL